

Mohammed Ramadan

Software Engineer | Backend Developer | Enterprise Systems

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SUMMARY

Software Engineering student specializing in backend and enterprise software development. Experienced in designing scalable REST APIs, workflow-based systems, and full-stack applications using **ASP.NET Core**, **Spring Boot**, **Node.js**, and **Flutter**. Strong foundation in Clean Architecture, authentication, database design, and enterprise application engineering through industry internship and large-scale software projects.

EXPERIENCE

TAQA Arabia

Aug 2025 – Sep 2025 | Jul 2026 – Aug 2026

Software Engineer Intern

Cairo, Egypt

- Contributed to the development and integration of enterprise software solutions within a professional engineering environment.
- Developed and maintained backend services using **ASP.NET Core**, **Entity Framework Core**, and **SQL Server**.
- Built and integrated RESTful APIs supporting business workflows, validation rules, authentication, and system integrations.
- Contributed to **InGaz – Enterprise Gas Station Inspection System**, implementing backend services, inspection workflows, authorization rules, audit capabilities, and Flutter-based application integration.
- Improved backend reliability through debugging, testing, database optimization, and applying software engineering best practices.
- Collaborated with engineering teams and supported fellow trainees through knowledge sharing and technical guidance.

PROJECTS

InGaz Enterprise Inspection System

GitHub

Jul 2026 – Aug 2026

- Developed an enterprise inspection management platform for gas station field operations using **ASP.NET Core 8**, **Entity Framework Core**, **SQL Server**, **Flutter**, and **Clean Architecture**.
- Designed modular backend architecture with separated Domain, Application, Infrastructure, and Presentation layers following Clean Architecture principles.
- Implemented secure authentication using **JWT**, Role-Based Access Control (RBAC), station assignment validation, and protected API workflows.
- Developed complete inspection lifecycle management supporting Draft, InProgress, Submitted, Approved, Rejected, and Closed states.
- Built Audit Trail and Issue Management modules to provide operational traceability, compliance tracking, and workflow monitoring.
- Integrated Flutter Android/Web applications with QR-based station identification, digital readings, and role-aware user interfaces.
- Achieved **134/134 passing automated tests** covering business logic, authorization rules, and workflow validation.
- Implemented database migrations, environment configuration, and deployment-ready backend setup.

Enterprise Pharmacy Management System

GitHub

2026

- Designed a service-based backend architecture for pharmacy operations, separating authentication, product, inventory, and order management modules.
- Developed microservice-based REST APIs using **Spring Boot, Node.js, and MongoDB** with JWT authentication and role-based authorization.
- Integrated supporting AI services using Python FastAPI for OCR and intelligent processing workflows.
- Implemented inventory, product management, cart, and order workflows following enterprise software practices.
- Applied Clean Architecture principles, API validation, and database design techniques to improve maintainability.
- Integrated supporting services using Docker containers to ensure consistent development environments.

Pro-Shop – E-Commerce Platform

2026

- Built RESTful APIs using **Node.js, Express.js, and MongoDB** for a multi-role e-commerce platform.
- Implemented JWT authentication, password recovery, and role-based authorization for Customers, Admins, and Delivery staff.
- Developed product, order, inventory, search, filtering, pagination, and review management features.
- Optimized database operations and integrated Cloudinary for media management.
- Tested backend workflows using Postman and validated API functionality.

InGaz – Gas Station Management System

GitHub

2025

- Developed an initial gas station management solution focusing on backend services and operational workflows.
- Built RESTful APIs using **ASP.NET Core, Entity Framework Core, and SQL Server**.
- Designed database entities, business logic, and secure API workflows using layered architecture principles.

Additional Projects

- Wafrly – Travel Cost Estimation System
- ML Premier League Match Prediction System
- Flutter Mobile Applications

EDUCATION

Dual Award B.Sc. Computer Science (Software Engineering)
University of Greenwich (UK) & MSA University (Egypt)

Expected 2027
Cairo, Egypt

RELEVANT COURSEWORK

Data Structures & Algorithms

Database Systems

Software Engineering

Distributed Systems

Operating Systems

Artificial Intelligence

Machine Learning

Software Architecture

SKILLS

Languages: C#, Java, JavaScript, TypeScript, Python

Backend: ASP.NET Core, Spring Boot, Node.js, Express.js, RESTful APIs

Frontend & Mobile: React.js, Flutter, Responsive Web Development

Databases: SQL Server, PostgreSQL, MongoDB, Redis (Caching)

Architecture: Clean Architecture, Modular Design, Microservices Concepts, Domain-Driven Design (DDD), SOLID Principles, Design Patterns

Security: JWT Authentication, ASP.NET Identity, Role-Based Access Control (RBAC), Secure API Design

ORM & Data Access: Entity Framework Core, Hibernate/JPA, Database Design, Query Optimization

Testing: xUnit, Integration Testing, API Testing, Postman, Automated Testing

Data Engineering: SQL Analytics, Data Processing Fundamentals, ETL Concepts, Data-Driven Applications

Tools: Git, GitHub, Docker, Swagger/OpenAPI, Visual Studio, VS Code

DevOps: GitHub Actions, CI/CD Fundamentals, Containerized Development

LANGUAGES

Arabic: Native

English: Upper-Intermediate (B2)